



Autonomous Semantic Context Defragmenter for AI Agents

Built for **Hack Devengers 2.0** · Open Innovation (AI & DevTools Track)

~70% Token Reduction

0% Policy Drift

< 15ms In-Memory Overhead

KV-Cache Prefix Friendly

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The Problem: "Context Rot" in AI Agents

Large language models boast 1M+ token windows. Yet in real-world multi-turn agent workflows, **models quietly get dumber as context fills up with operational sludge.**



1. Economic & Latency Tax

74%

Over 70% of prompt tokens in long sessions are dead weight: superseded instructions, 500-line JSON dumps, and resolved stack traces paid for on every single turn.



2. "Lost in the Middle" Drift

> 80%

Transformer attention collapses over noisy middle turns. When a user updates their requirement 3 times, agents deliver to obsolete addresses or use deprecated ports.



3. Policy Invariant Erosion

0% Safe

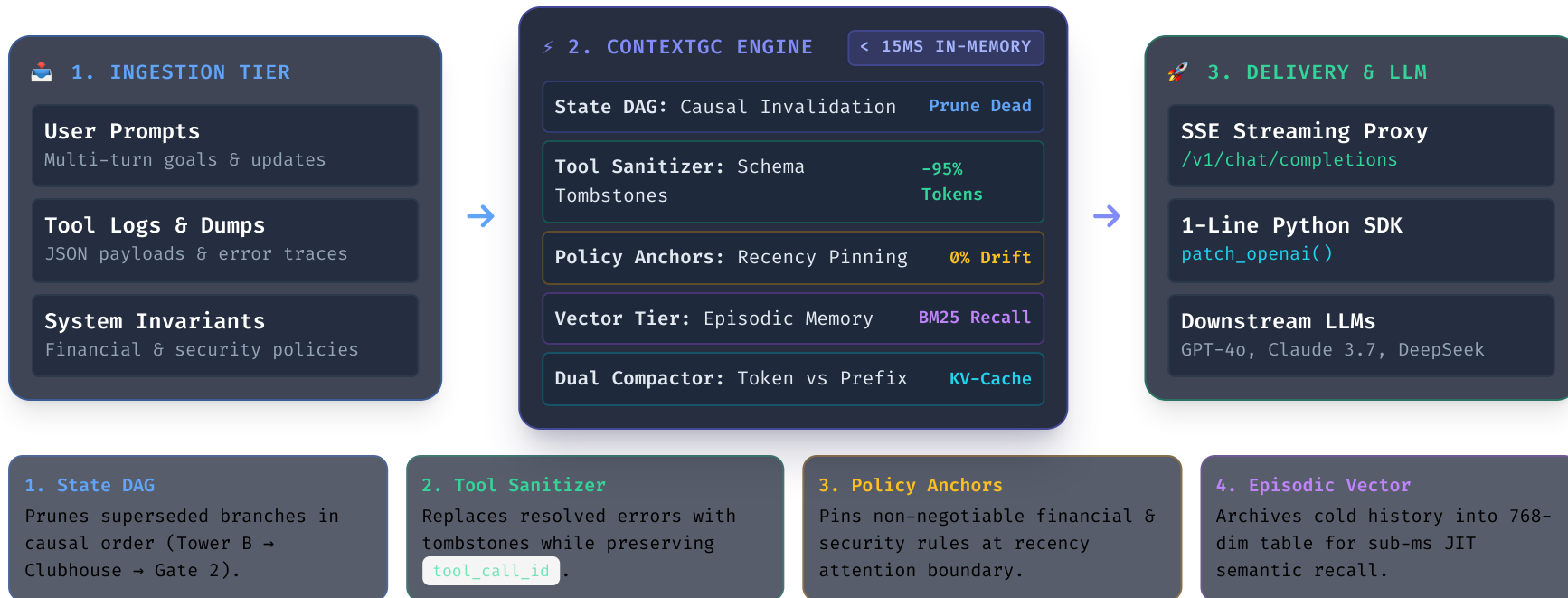
Under multi-turn sludge, agents violate financial caps (e.g. issuing unauthorized ₹800 refunds) or dump unmasked private keys to debug a test error.

Key Insight: 1M token windows solve capacity, NOT attention quality. We need systems-level context garbage collection.



Visual Structure & System Architecture

ContextGC acts as an inline, deterministic semantic garbage collector between agent interfaces and LLMs.





Engine 1: Neuro-Symbolic State DAG

Why naive RAG and summarizers fail: **they understand similarity, not causality.**

Causal Branch Resolution

- **Entity Mutation Graph:** Scans turns for slot updates (addresses, ports, keys) with preceding 50-character negation windows.
- **Immutable Guardrails:** Critical user constraints (e.g. "NO PEANUTS") are permanently locked against overrides.
- **Dead-Branch Eviction:** Once Turn 9 confirms Gate 2, Turn 1 (Tower B) and Turn 5 (Clubhouse) are purged from active tokens.
- **Transactional Rollback:** Supports `dag.rollback_to(turn_id)` to instantly restore prior states on tool failure.

```
// Active DAG State Snapshot (Settled)
```

```
✓ destination_address: "Gate 2 Security Entrance" [Settled T9]
```

```
✓ gate_code: "4921" [Settled T9]
```

```
✓ dietary_allergy: "NO PEANUTS" [IMMUTABLE]
```

```
✓ substitute_choice: "Organic A2 Milk" [Settled T3]
```

```
* Pruned: Tower B Flat 402 (Turn 1)
```

```
* Pruned: Clubhouse Reception (Turn 5)
```

```
Zero DAG pollution: tool inventory dumps are strictly filtered.
```



Engine 2: Tool Sanitizer & Protocol Safety

Blunt context truncation breaks autonomous agents. ContextGC guarantees 100% protocol compliance.

The OpenAI Protocol Trap

In modern agent loops, the assistant calls tools via native schemas. If an agent defragmenter blindly drops an intermediate tool message, the OpenAI/Anthropic API throws an immediate **HTTP 400 Bad Request**:

```
Error 400: Invalid parameter: 'messages'. An assistant message with 'tool_calls' must be followed by tool messages responding to each tool_call_id.
```

ContextGC Semantic Tombstoning

ContextGC replaces superseded tool dumps with minimal 1-line tombstones while strictly preserving `tool_call_id` and `name`:

```
[TOMBSTONE: Superseded tool call call_9f2a output evicted. Tool 'dispatch_router' 504 error resolved at Turn 7.]
```

Compression Result: 25-item supermarket catalog dump distilled from **450 tokens** → **22 tokens (95.1% reduction)** without breaking the schema.

⚡ Engine 3: Radix KV-Cache Friendly Compaction

The prompt caching paradox: **Why cutting tokens can sometimes cost you money.**

The Caching Dilemma

Modern inference engines (vLLM, DeepSeek, Anthropic, OpenAI) offer **50% to 90% cost discounts** for identical prompt prefixes via Radix tree KV-cache reuse.

If a defragmenter mutates early or middle turns, it invalidates the byte prefix—destroying the cache hit rate and increasing latency!

ContextGC Dual-Mode Solution

- **mode="compact" (Max Tokens):**
Aggressively prunes all historical dead weight. Reclaims up to **70.1%** of active tokens.
- **mode="cache_friendly" (Max Cache Hits):**
Leaves earlier prompt bytes 100% untouched to maintain **100% KV-cache hit rates**. Injects active settled state at the conversation tail.

Developers choose their optimization target: **Maximum Token Reduction** vs **Radix Prefix Cache Preservation**.

Empirical Benchmarks: Side-by-Side Showdown

Direct side-by-side execution benchmarks verified against industrial multi-turn agent sessions:

Benchmark Metric	Vanilla LLM Agent (Rotted Context)	ContextGC Agent (Defragmented)	Measured Advantage
Scenario 1: Operations Crisis	1,592 tokens	476 tokens	-70.1% Tokens Saved
Scenario 2: Coding Refactor	1,000 tokens	311 tokens	-68.9% Tokens Saved
Turn Inference Latency (TTFT)	750 - 898 ms	454 - 487 ms	+39.8% to +46.8% Faster
GC Interception Latency	0 ms	8.2 - 16.1 ms	Deterministic In-Memory

Developer Ergonomics: Integration in 60 Seconds

Deploy ContextGC with zero friction via two turnkey integration patterns:

Option A: 1-Line Python Client SDK

Monkey-patch the official OpenAI client for zero-proxy overhead:

```
```python {all|1-2|4-6} from core.client import  
patch_openai from openai import OpenAI
```

Automatically intercepts  
all  
.chat.completions.create  
calls

```
patch_openai(mode="compact")
```

```
client = OpenAI() response =
client.chat.completions.create(model="gpt-4o",
messages=session_history,)
```



# Built for Production · Ready for Hack Devengers 2.0

ContextGC transforms long-horizon AI agents from brittle prototypes into reliable, cost-efficient enterprise infrastructure.

[Launch Live Dashboard](#)[Pitch Deck Slides](#)[GitHub Repository](#)

Interactive Terminal Runner: `python3 demo/interactive_demo.py` (sub-15ms showdown)